Experiment No. 22 – Closing Morphological Operation

1. Aim

To implement the Closing technique as a Morphological Operation using OpenCV to fill small holes and close gaps in foreground regions.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Image loading failed")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Convert to binary image
_, binary = cv2.threshold(
    gray, 127, 255, cv2.THRESH_BINARY
)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply closing operation
closing = cv2.morphologyEx(
    binary, cv2.MORPH_CLOSE, kernel
)

# Display output
plt.figure(figsize=(12, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(binary, cmap="gray")
plt.title("Binary Image")
plt.axis("off")
```

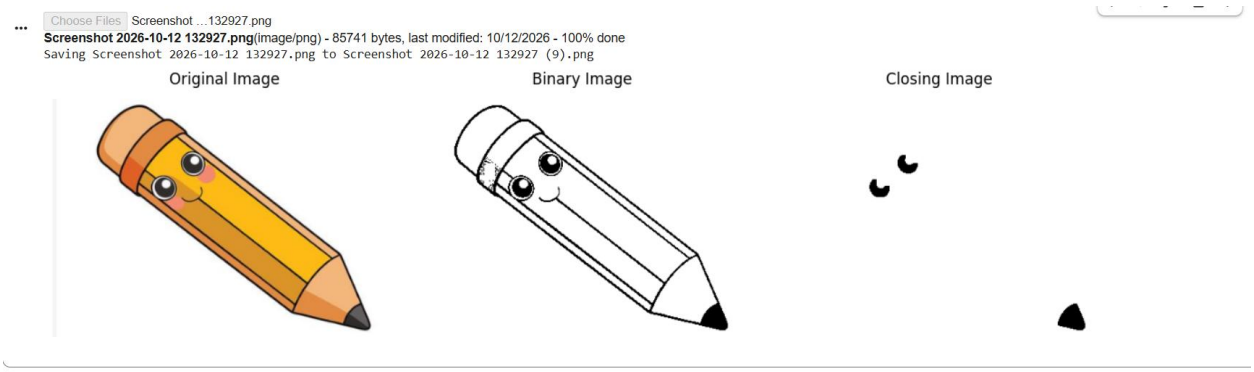
```
plt.subplot(1, 3, 3)
plt.imshow(closing, cmap="gray")
plt.title("Closing Image")
plt.axis("off")


plt.tight_layout()
plt.show()
```

3. Expected Output

The output shows small holes and gaps in the white foreground regions being filled.

4. Output



The closing operation is successfully applied to fill small holes and connect nearby foreground regions.

5. Result

Thus, the Closing Morphological Operation was successfully implemented using Python and OpenCV.